airtel

SYNC

# Udyam Digital

A mobile-friendly website and WhatsApp journey that takes a small business from one sentence to a booked installation, in its own language.



**Team NATPAT · TIET, Patiala**

Airtel SYNC 2026 · Digital Track

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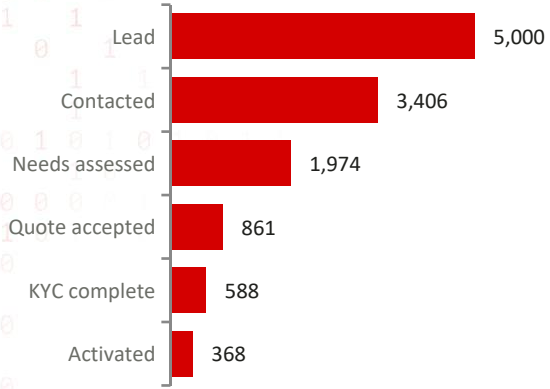
Demo video (2:58) · Code on GitHub · Architecture and privacy note

Platform Partner unstop

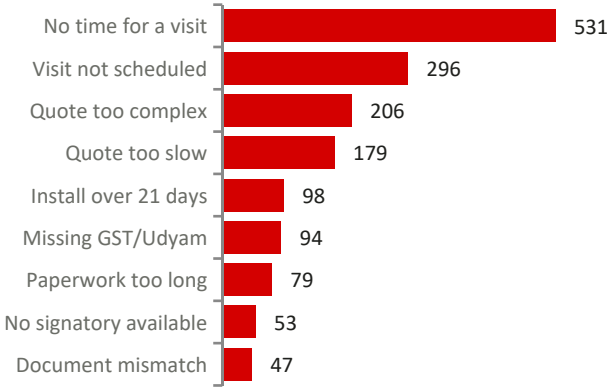
# 7 in 100 leads activate, and a third of the loss is addressable

One quarter of synthetic SMB leads (Apr–Jun 2026). The steepest drop is from needs assessed to quote accepted.

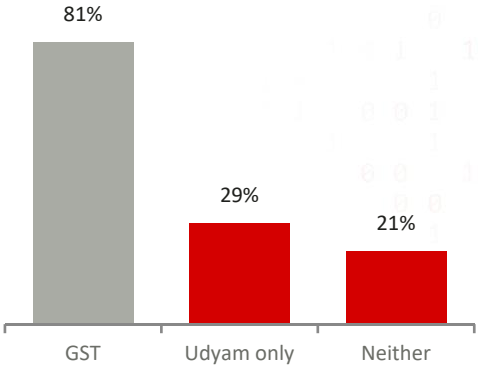
The funnel




Addressable drop reasons: 1,583 leads




Clear KYC after accepting a quote




**17.6 days**
lead to live; 12.9 of them hand-offs

**3 calls**
per customer, chasing status

**34%**
of 4,632 drops (1,583) cite reasons a digital journey can address

**3 in 4**
owners without GST fail KYC after saying yes



Source: smb_leads.csv (5,000 synthetic leads; 368 activations; ₹55.4 lakh acquisition spend). KYC: has_gstin, has_udyam.

Platform Partner unstop

# Today’s buying journeys leave gaps; we close them in one sitting

The three ways a small business buys connectivity today, compared at each step of the journey.

| Step | Accel field sales today | Digital-first, single product | Local reseller, many vendors | Udyam Digital |
|---|---|---|---|---|
| Find the right product | A visit; the executive pitches what is on target | Self-serve, one product at a time | Whatever the shop stocks | **Advisor in 10 languages, web or WhatsApp** |
| What you can buy | Products sold one at a time | A single product | Pieces from different vendors | **A whole bundle, a reason for every line** |
| Price | Quote in 3–5 days, PDF with product codes | Instant, for that product | Informal | **Instant monthly price, add-ons to toggle** |
| Verification | Papers collected in person; 3.8 days | Online sign-up | Varies | **Registry + documents read on the phone: minutes** |
| Owners without GST | 1 in 4 clears KYC | Varies | Varies | **Free Udyam, guided; or Aadhaar e-KYC + shop proof** |
| Status and support | 3 status calls; billed and supported per product | Per product | No single bill or support number | **Dated tracker on WhatsApp; one bill** |

**Where we win:** the bundle, one bill, the owner’s language, and a route for the 29% of leads with no GSTIN. A partner steps in when software cannot close.

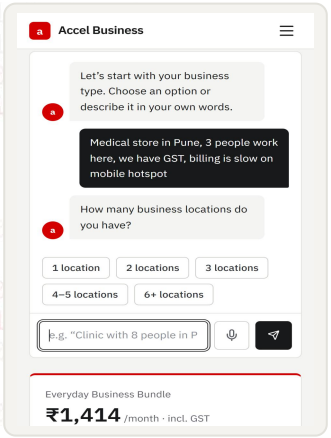
**Where they win, and our answer:** a single-product sign-up is quicker for one SIM, so owners can drop any extra line from the quote. Resellers own local trust, so we recruit them as paid partners.



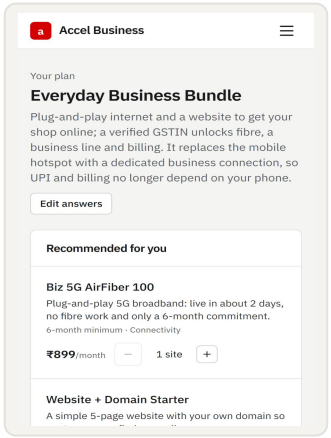
Sources: the case brief and smb_leads.csv. "Varies": not described in the brief, so we make no claim.

# One sentence to a booked installation, without an initial sales visit

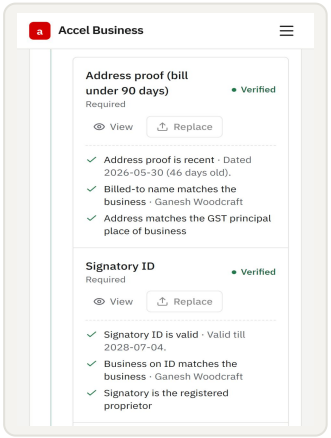
**Fixes:** visits, slow quotes and status chasing (slide 2). Screens from the working build.



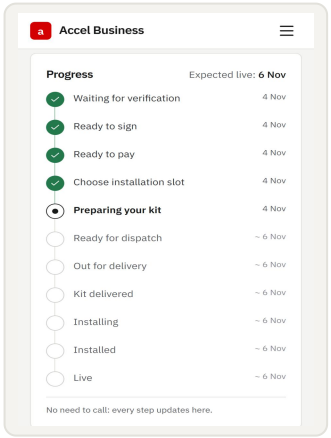

**1 Describe the business**



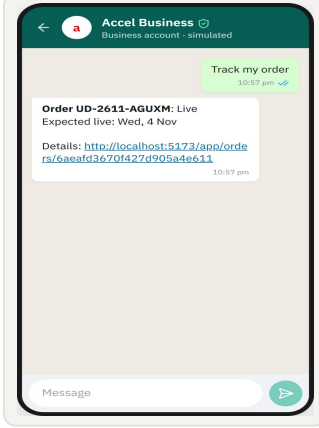

**2 See the price**




**3 Verify once**




**4 Track to go-live**




**5 Or stay on WhatsApp**

| Owner | API | Data | Connections |
|---|---|---|---|
| BUILT | BUILT | BUILT | BUILT |
| Web app (React) or WhatsApp, 10 languages | Advisor, pricing engine, Smart KYC rules, orders, lead model | MongoDB; private Cloudflare R2 document store | GST and Udyam registries, Aadhaar e-KYC, UPI, WhatsApp, Claude |



The demo runs on the case's synthetic data, with payments in test mode. Full map: architecture.png.

# Clean applications verified in minutes, including owners with no GST

**Fixes:** paperwork drops, a 3.8-day KYC stage, and 3 in 4 owners without GST failing KYC (slide 2).

| | |
|---|---|
| KYC today | **3.8 days** |
| **Automated checks** | **Minutes** |

Flagged cases: reviewed within 4 hours

1. **Consent**
   one purpose, 30 minutes
2. **Registry lookup**
   GSTIN or Udyam number
3. **Bill and ID read**
   on the phone, ~1.5 s a page
4. **Decision**
   verified, or exactly what to fix

| Issues identified in sample applications | Caught | The owner is told |
|---|---|---|
| GST registration cancelled | ✓ | Use an active GSTIN, or Udyam |
| Certificate photo too blurred | ✓ | Retake a sharper photo |
| GSTIN altered or mistyped (2) | ✓ | The registry's GSTIN |
| Legal name differs from registry | ✓ | Fix it, or ask a reviewer |
| Bill 257 days old (limit 90) | ✓ | Upload a recent bill |
| 6 clean applicants | ✓ | **Verified; reused for a year** |

## Owners without GST

**Get Udyam free, guided:** details pre-filled; a reminder after a day, a partner's call after three.

**Or verify without registering:** Aadhaar e-KYC by OTP, plus a bill and a shop licence or shop photo.

**Then:** everything that does not need GST can be bought.

## DPDP by design

- Consent first: one purpose, 30 minutes
- Aadhaar never stored; a typo is caught before any OTP
- Files private; every view audited



Sample applicants K01–K12 (47 images); answer key not shared with teams. Details: privacy_note.pdf.

# Offices and multi-site first; partners and a model take the rest

Priority follows conversion and how fast revenue covers the cost to win. Micro retail stays digital-only.

| Segment | Activate | Cost to win | Bundle /mo | Revenue months | Launch offer and channel |
|---|---|---|---|---|---|
| 1 · Professional offices | 11.1% | ₹10,695 | ₹6,796 | 1.6 mo | Fibre 300, 2 lines, email, backup · self-serve |
| 2 · Small multi-site | 11.0% | ₹15,614 | ₹31,233 | 0.5 mo | Fibre + failover + CCTV per site · digital + partner survey |
| 3 · Small manufacturers | 12.8% | ₹15,324 | ₹19,980 | 0.8 mo | Leased line, firewall, IoT · partner-led, digital KYC |
| 4 · Micro retail | 4.1% | ₹20,007 | ₹2,505 | 8.0 mo | Fibre 100 or 5G AirFiber + POS · WhatsApp only |

**Revenue months = cost to win ÷ monthly bundle revenue** (the pack has no margins), e.g. ₹10,695 ÷ ₹6,796 = 1.6.

**Digital advisor:** Every segment self-serves; WhatsApp is the cheapest channel (₹4,026 per activation).

**Channel partners:** Own the live queue: failed payments, KYC help, callbacks. Paid per activation, so they gain.

**Field force:** Site surveys for multi-site and manufacturers only (₹24,335 per activation).

## The lead model

**Predicts:** how likely a stalled lead is to activate.

**Inputs:** lead source, segment, city tier, GSTIN, Udyam, team size, sites, spend, current internet.

**Result:** logistic regression; AUC 0.705 on held-out leads; the top 10% activate at 16% vs 8.1%.

**Changes the work:** partners call the best leads first, each shown with its reasons.

**Fairness:** no names, phone numbers or language; city tier only, never the exact city. Checked quarterly by language and city tier.



Source: smb_leads.csv (cost to win = acquisition spend ÷ activations). Model: 3,749 leads to train, 1,251 to test.

Platform Partner unstop

# +237 activations a quarter at 39% lower acquisition cost

Base case: same leads and spend; the journey recovers 15% of the 1,583 digitally addressable drops.

| Metric | Today | Month 12 | Why reachable |
|---|---|---|---|
| Lead-to-activation | 7.4% | 12% | Recover 15% of addressable drops |
| Days to activate | 17.6 | ≤ 7 | 12.9 days of hand-offs removed |
| Cost to win | ₹15,053 | ≤ ₹9,200 | Same spend, more activations |
| Status calls | 3.0 | ≤ 0.5 | Dated tracker and WhatsApp |
| 6-month churn | 15.2% | ≤ 10% | One-call customers: 5.9% |

## Rollout

**Q1 pilot:** offices and retail in 2 tier-1 cities; web, WhatsApp, partners.
**Q2–Q3 scale:** multi-site; GSTN API, DigiLocker, e-sign, live UPI.
**Q4 national:** manufacturers with partners; tier-2 cities.

| Drops recovered | 10% | 15% (base) | 30% |
|---|---|---|---|
| Activations a quarter | 526 | **605** | 843 |
| Cost to win | ₹10,532 | **₹9,156** | ₹6,571 |

**₹1.96 Cr** annualised revenue from one quarter's extra activations (237 × ₹6,900 × 12), before churn. **+43** more if 60% of owners without GST clear KYC (25% today): 190 × 60% × 30/47 = 73, vs 30. About 7 overlap the base case.

## Risks and mitigations

**Recovery below 15%:** even 10% adds 158 activations and cuts cost to win by 30%.
**Fraud or poor scans:** checksum and registry checks; reviewer below 60% confidence.
**Channel conflict:** partners are paid on assisted activations.
**AI mistakes:** the catalogue, not the model, sets products and prices.
**DPDP and telecom KYC:** consent, audit, erasure built; licensed e-KYC partner.



Scenario on smb_leads.csv: ₹55.4 lakh acquisition spend held constant; 368 activations a quarter today.